Stochastic Sign Descent Methods: New Algorithms and Better Theory

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The Problem

smooth & non-convex | # machines/devices | smooth & non-convex

$$\min_{x\in\mathbb{R}^d}\left\{ f(x) \overset{\text{def}}{=} \frac{1}{M}\sum_{n=1}^{M} f_n(x)\right\}$$

model parameters | empirical risk/loss | local training data | local loss function $f_n(x) = \mathbb{E}_{\xi\sim\mathcal{D}_n}[f_\xi(x)]$

Assumptions

Success Probability Bounds (SPB). For any $x\in\mathbb{R}^d$, we have (not necessarily unbiased) estimator $\hat{g}(x)$ of gradient $g(x) \overset{\text{def}}{=} \nabla f(x)$ such that

$$\rho_i(x) \overset{\text{def}}{=} \Pr(\operatorname{sign}\hat{g}_i(x) = \operatorname{sign} g_i(x)) > \tfrac{1}{2}, \quad \text{if} \quad g_i(x)\neq 0 \qquad (1)$$

for all $x\in\mathbb{R}^d$ and all $i\in\{1,2,\dots,d\}$.

Bounded Variance (BV). For any $x\in\mathbb{R}^d$, each node n has access to an unbiased estimator $\hat{g}^n(x)$ with bounded variance $(\sigma^n)^2\geq 0$, namely

$$\mathbb{E}[\hat{g}^n(x)] = \nabla f_n(x), \quad \mathbb{E}\left[\|\hat{g}^n(x)-\nabla f_n(x)\|^2\right] \leq (\sigma^n)^2. \qquad (2)$$

Smoothness Constants L_i and L^n correspond to coordinate i for function $f(x)$ and node n for function $f_n(x)$ respectively.

Sufficient Conditions for SPB

Fix $x\in\mathbb{R}^d$ and let $\hat{g}(x)$ be an unbiased estimator of the gradient $g(x)$.

Lemma 1. If each coordinate $\hat{g}_i(x)$ has a unimodal and symmetric distribution with variance $\sigma_i^2(x) \overset{\text{def}}{=} \mathbb{E}[|\hat{g}_i(x)-g_i(x)|^2]$, $1\leq i\leq d$, then

$$\rho_i(x) \geq \tfrac{1}{2} + \tfrac{1}{2}\tfrac{|g_i(x)|}{|g_i(x)|+\sqrt{3}\sigma_i(x)} > \tfrac{1}{2} \quad \text{if} \quad g_i(x)\neq 0.$$

Lemma 2. If variances $\sigma_i^2(x)\leq c_i g_i^2(x)$ are bounded for some constants $c_i\geq 0$ and $\tau > 2\max_i c_i$ mini-batch of stochastic gradients are used, then

$$\rho_i(x) \geq 1-\tfrac{c_i}{\tau} > \tfrac{1}{2}, \quad \text{if} \quad g_i(x)\neq 0.$$

Lemma 3. If $\sigma_i^2(x)$ is the variance and $\nu_i^3(x) \overset{\text{def}}{=} \mathbb{E}[|\hat{g}_i(x)-g_i(x)|^3]$ is the 3th central moment of $\hat{g}_i(x)$, $1\leq i\leq d$, then

$$\rho_i(x) > \tfrac{1}{2} \quad \text{if mini-batch size } \tau > 2\min\left(\tfrac{\sigma_i^2(x)}{g_i^2(x)}, \tfrac{\nu_i^3(x)}{|g_i(x)|\sigma_i^2(x)}\right).$$

A New "Norm" for Measuring the Gradients

Definition 1 (ρ_M-norm). *Let $l = \lfloor \frac{M+1}{2}\rfloor$ and I be the regularized incomplete beta function. Define the ρ_M-norm of gradient $g(x)$ via*

$$\|g(x)\|_{\rho_M} = \textstyle\sum_{i=1}^d \left(2I(\rho_i(x); l, l)-1\right)|g_i(x)|.$$

We show $\|g(x)\|_{\rho_M} \to \|g(x)\|_1$ exponentially fast as $M\to\infty$, i.e.,

$$\left(1-\exp(-(2\rho(x)-1)^2 l)\right)\|g(x)\|_1 \leq \|g(x)\|_{\rho_M} \leq \|g(x)\|_1,$$

where $\rho(x) = \min_{1\leq i\leq d}\rho_i(x) > \frac{1}{2}$.

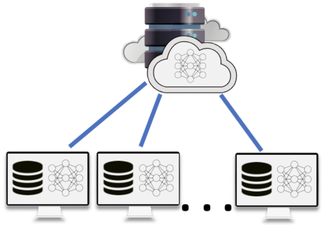
Single Machine Training

Data to train → Centralized

Compression → Standard sign

Aggregation → NA

Parallel Training

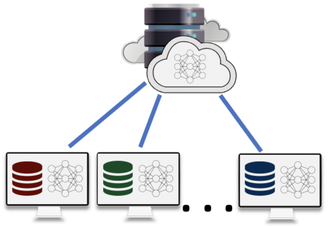


Data to train → Shared or IID

Compression → Standard sign

Aggregation → Majority vote

Distributed Training



Data to train → Heterogeneous or non-IID

Compression → Stochastic sign

Aggregation → Arithmetic mean

Single Machine Training ($M=1$)

Algorithm 1 SIGNSGD [1,2]

Input: step size γ_k, current point x_k
$\hat{g}_k \leftarrow \text{StochasticGradient}(f, x_k)$
$x_{k+1} = x_k - \gamma \operatorname{sign}\hat{g}_k$

Parallel Training

Algorithm 2 PARALLEL SIGNSGD [1,2]

Input: step size γ, current point x_k, # of nodes M
on each node n
  $\hat{g}_k^n \leftarrow \text{StochasticGradient}(f, x_k)$
on server
  get $\operatorname{sign}\hat{g}_k^n$ **from** all nodes
  send $\operatorname{sign}\left[\sum_{n=1}^M \operatorname{sign}\hat{g}_k^n\right]$ **to** all nodes
on each node n
  $x_{k+1} = x_k - \gamma \operatorname{sign}\left[\sum_{n=1}^M \operatorname{sign}\hat{g}_k^n\right]$

If **SPB** (1) holds, then

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|\nabla f(x_k)\|_{\rho_M} \leq \frac{f(x_0)-f^*}{\gamma K} + \gamma\frac{d\bar{L}}{2}$$

total # iterations | lower bound $f^*\leq f(x), \forall x\in\mathbb{R}^d$ | $\bar{L} = \frac{1}{d}\sum_{i=1}^d L_i$

Distributed Training

Algorithm 3 STOCHASTIC SIGN DESCENT WITH MOMENTUM (SSDM) **(new)**

Input: step size parameter γ, momentum parameter β, # of nodes M
Initialize: $x_0\in\mathbb{R}^d$, $m_{-1}^n = \hat{g}_0^n$
for $k = 0,1,\dots,K-1$ **do**
  on each node n
    $\hat{g}_k^n \leftarrow \text{StochasticGradient}(f_n, x_k)$ Local sub-sampling
    $m_k^n = \beta m_{k-1}^n + (1-\beta)\hat{g}_k^n$ Update the momentum
    send $s_k^n := \widetilde{\operatorname{sign}}\, m_k^n$ **to** the server
  on server
    send $s_k := \sum_{n=1}^M s_k^n$ **to** all nodes
  on each node n
    $x_{k+1} = x_k - \frac{\gamma}{M}s_k$ Main step: Update the global model
end for

If **BV** (2) holds, $\beta = 1-\frac{1}{\sqrt{K}}$ and $\gamma = \frac{1}{K^{3/4}}$, then

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|\nabla f(x_k)\| \leq \frac{16}{K^{1/4}}\left[\delta_f + \tilde{\sigma} + \tilde{L}\sqrt{d} + \frac{\tilde{L}d}{K}\right]$$

l^2 norm | $\delta_f = f(x_0)-f^*$ | $\tilde{\sigma} = \frac{1}{M}\sum_{n=1}^M \sigma^n$ | $\tilde{L} = \frac{1}{M}\sum_{n=1}^M L^n$

Stochastic Sign Compressor (new)

Definition 2. *Define the stochastic sign operator* $\widetilde{\operatorname{sign}}: \mathbb{R}^d\to\mathbb{R}^d$ *via*

$$\left(\widetilde{\operatorname{sign}}\, g\right)_i = \begin{cases} +1, & \text{with probability } \frac{1}{2}+\frac{1}{2}\frac{g_i}{\|g\|} \\ -1, & \text{with probability } \frac{1}{2}-\frac{1}{2}\frac{g_i}{\|g\|}\end{cases}$$

for $i\in\{1,2,\dots,d\}$ and $\widetilde{\operatorname{sign}}\,\mathbf{0} = \mathbf{0}$ with probability 1.

(i) Stochastic $\widetilde{\operatorname{sign}}$ compressor can be viewed as a noisy version of standard deterministic sign compressor [3].

(ii) Next, it satisfies **SPB** (1) automatically, i.e., for all $i\in\{1,2,\dots,d\}$

$$\Pr((\widetilde{\operatorname{sign}}\, g)_i = \operatorname{sign} g_i) = \tfrac{1}{2}+\tfrac{1}{2}\tfrac{|g_i|}{\|g\|} > \tfrac{1}{2}, \quad \text{if} \quad g_i\neq 0.$$

(iii) Lastly, unlike the deterministic sign operator, it is unbiased with scaling factor $\|g\|$, namely $\mathbb{E}[\|g\|\,\widetilde{\operatorname{sign}}\, g] = g$ for any vector $g\in\mathbb{R}^d$.

Experiments

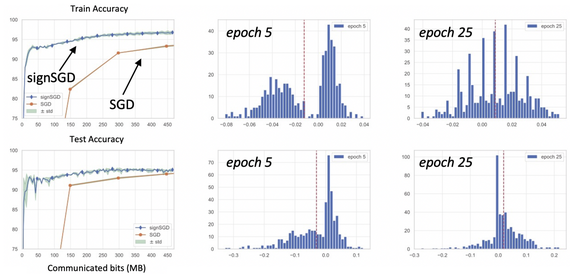


Figure 1: Convergence of signSGD and comparison with SGD on the MNIST dataset using the split batch construction strategy. The budget of gradient communication (MB) is fixed and the network is a single hidden layer FNN. We first tuned the constant step size over logarithmic scale $\{1, 0.1, 0.01, 0.001, 0.0001\}$ and then fine tuned it. First column shows train and test accuracies with mini-batch size 128 and averaged over 3 repetitions. We chose two weights (empirically, most of the network biases would work) and plotted histograms of stochastic gradients before epochs 5 and 25. Dashed red lines on histograms indicate the average values.

References

[1] Jeremy Bernstein, Yu-Xiang Wang, Kamyar Azizzadenesheli, and Animashree Anandkumar. *signSGD: Compressed optimisation for non-convex problems. ICML, 2018.*

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[3] Xiangyi Chen, Tiancong Chen, Haoran Sun, Zhiwei Steven Wu and Mingyi Hong. *Distributed Training with Heterogeneous Data: Bridging Median- and Mean-Based Algorithms. NeurIPS, 2020.*

[4] Mher Safaryan and Peter Richtárik. *Stochastic Sign Descent Methods: New Algorithms and Better Theory. ICML, 2021.*